

UNIT 1: Fundamentals of AI Products & Emerging AI Systems



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TOP 20 AI PRODUCTS SHAPING OUR FUTURE TRANSFORMING DAILY LIFE

amazon alexa



OpenAI
GPT-3



MATHia®

FAST FORWARD®
BY CARNEGIE LEARNING



Google Assistant

TESLA
AUTOPILOT

VIDEO



DOORBELL

SOFTBANK
ROBOTICS'
PEPPER



Betterment



IBM Watson



INTERCOM

zebra
MEDICAL VISION

intuit
mint.



TARANIS



LIVEPERSON®

BLUE RIVER
TECHNOLOGY
SEE & SPRAY

Image Credit - Social Media | Follow Us @ThinkWithNiche

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- Large Language Models
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Introduction to AI Products



- A software application or service that leverages artificial intelligence—such as machine learning, natural language processing, or computer vision.
- To automate tasks, analyze data, and simulate human intelligence, offering predictive, personalized, or generative capabilities They improve automatically using new data
- Examples: Chatbots, recommendation systems, voice assistants

Key Aspects of AI Products



- **Core Components:** Data (for training), Algorithms (for learning patterns), and Models (the resulting engine for decisions).
- **Types:**
 - **Native AI:** Products that cannot function without AI (e.g., ChatGPT, self-driving cars).
 - **AI-Enabled:** Existing products enhanced by AI features (e.g., personalized recommendations in Netflix, predictive text in Gmail).
- **Product Lifecycle:** Involves unique steps, including data acquisition, model training, validation, and continuous monitoring for accuracy.

AI Tool vs AI Product



Basis	AI Tool	AI Product
Definition	Software or platform used to develop AI models and applications	Complete application that uses AI to solve user problems
Users	Mainly used by developers, data scientists, and engineers	Used by general users or customers
Purpose	Helps in building, training, and testing AI systems	Provides AI-powered services to end users
Technical Knowledge	Requires programming and technical knowledge	Usually easy to use and does not require technical skills
Function	Provides algorithms, libraries, and frameworks	Integrates AI models with interface and features
Example	TensorFlow, PyTorch, Scikit-learn	ChatGPT, Netflix recommendation system, Siri

Components of AI Systems



- Data – Information used to train AI models
- Model – Algorithm that learns patterns
- Interface – Allows users to interact with system
- Feedback Loop – Improves model performance

Data



- **Training Data:** labeled datasets used to teach the model patterns, relationships, and features
- **Validation/Testing Data:** Separate data used to evaluate the model's performance on unseen, real-world scenarios.
- **Types:** Data can be structured (tables) or unstructured (text, images, video, sensor data).
- **Data Processing:** Before training, data must be cleaned, transformed, and formatted to remove noise and errors, often using tools like Apache Spark

Model



- **Algorithms:** The underlying logic (e.g., neural networks, decision trees) that defines how the model processes data.
- **Parameters:** Internal variables learned during training that are adjusted to minimize error.
- **Types:** Includes machine learning, deep learning and generative models.

Interface



- **Types:** Conversational agents (chatbots), voice user interfaces (VUIs), and graphical user interfaces (GUIs).
- **Function:** It serves as a bridge, allowing users to input data (prompts) and receive outputs.
- **Adaptive Design:** Modern interfaces are becoming hyper-personalized, adapting to user needs and context in real-time.

Feedback Loop



- **User Feedback:** Explicit signals (thumbs-up/down, ratings) or implicit signals (time spent on content, clicks).
- **Correction:** Correcting AI outputs or regenerating responses feeds back into the model to improve accuracy.
- **System Learning:** The process allows the model to learn from mistakes and improve its "reinforcement" in subsequent tasks

Overview of Generative AI

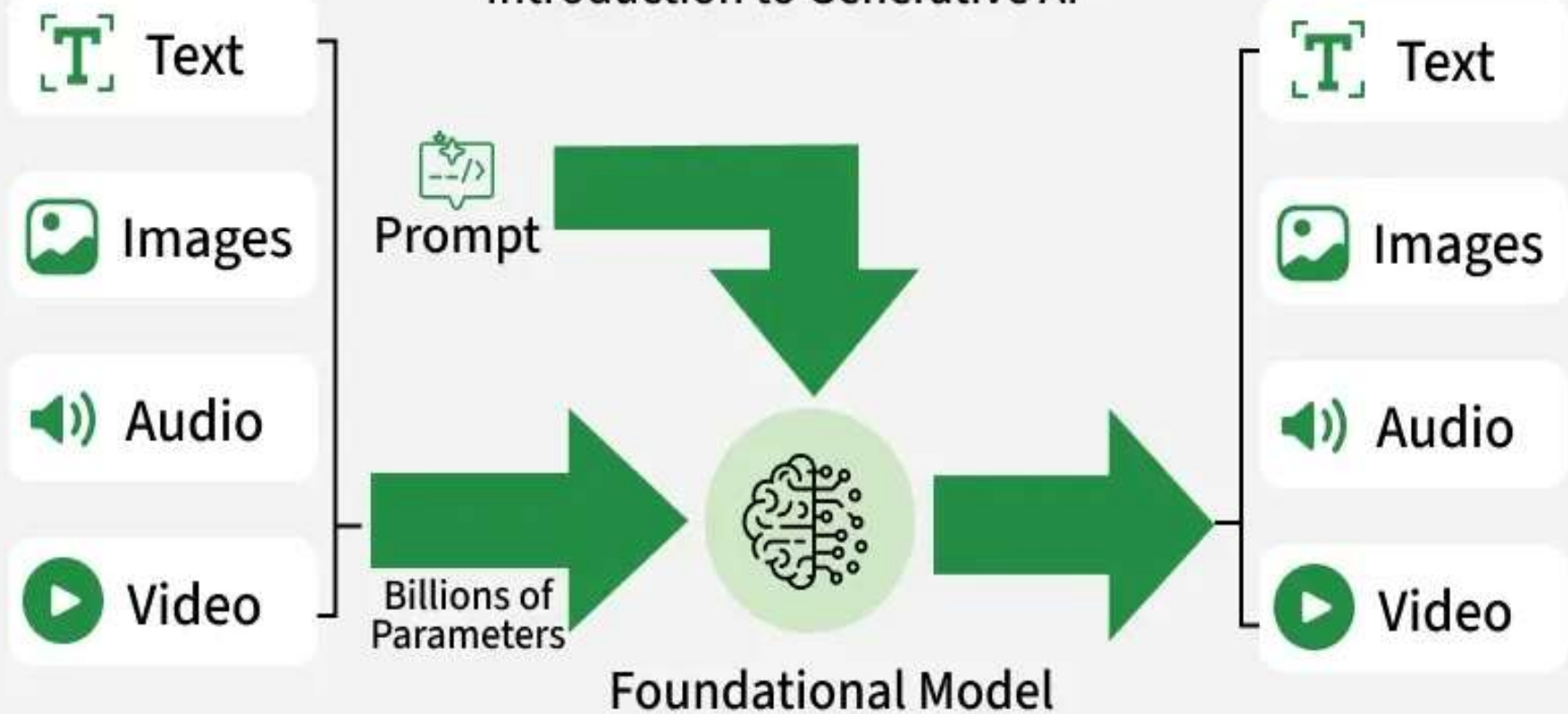


- **Type of Artificial Intelligence that can create new content automatically.**
- It can generate text, images, video, music or code

Generative AI



Introduction to Generative AI



Applications of Generative AI



- Content writing and blogging
- Image and graphic design
- Video creation
- Chatbots and virtual assistants
- Software code generation

Examples of Generative AI



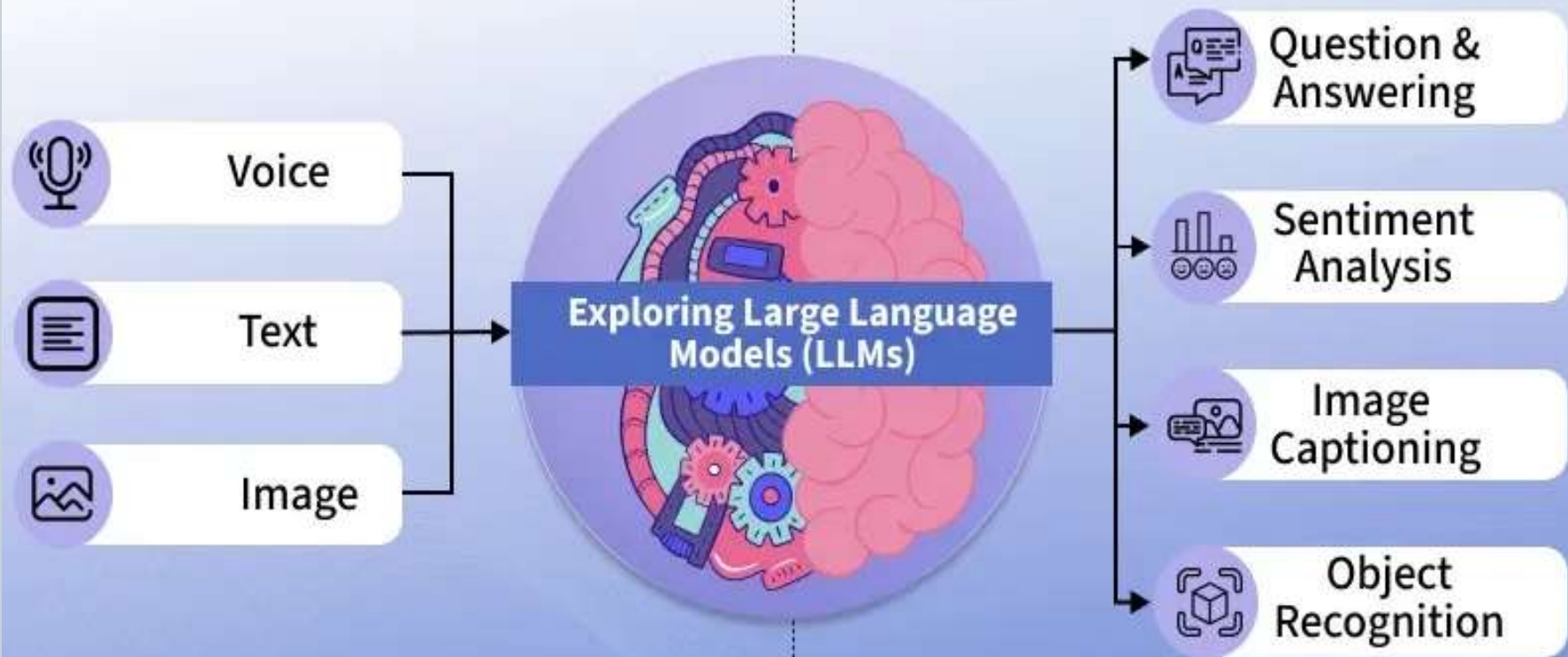
- **ChatGPT** – generates text and answers questions
- **DALL·E / Midjourney** – generates images from text prompts
- **GitHub Copilot** – generates programming code
- **AI music generators** – create music automatically

Large Language Models (LLMs)



- LLMs are trained on massive text datasets
- They understand and generate human language
- Used in chatbots, translation, writing assistance

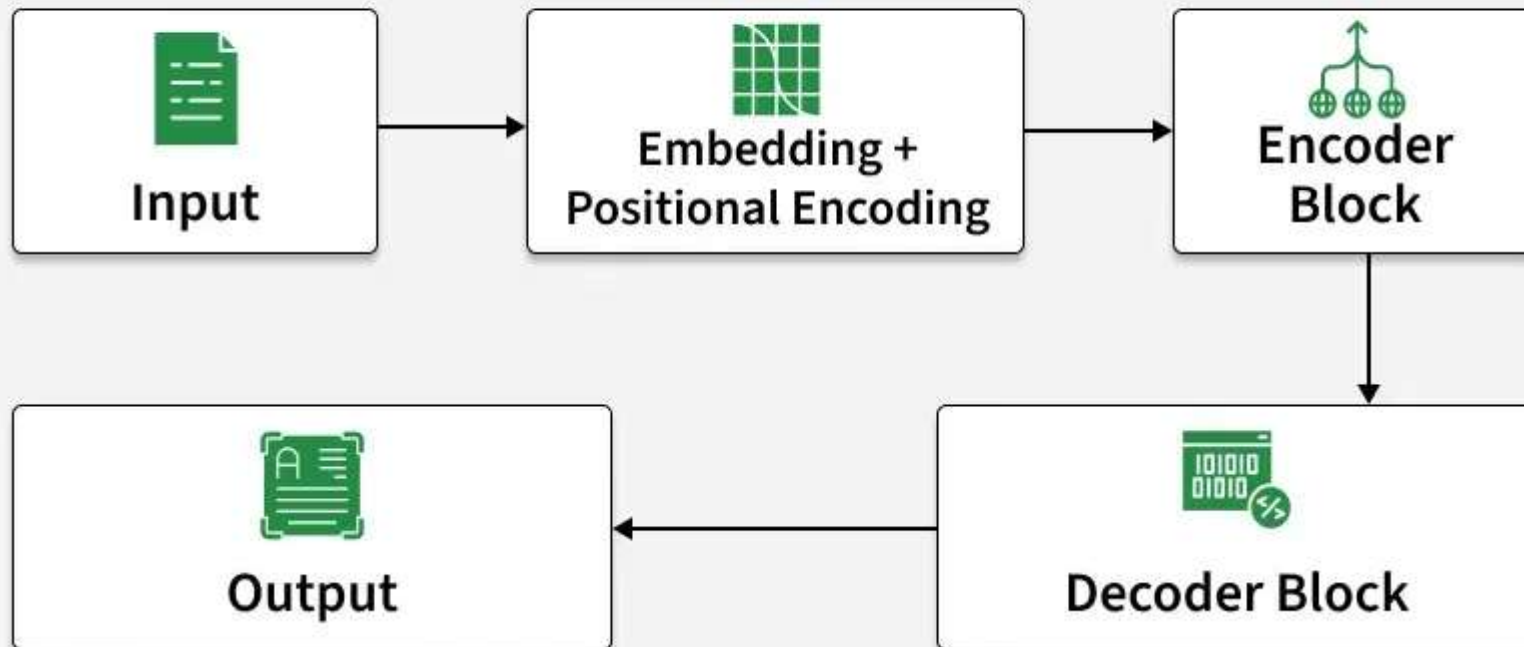
LLM



Working of LLM



Transformers in LLMs



Examples of LLMs



- **GPT models** (used in ChatGPT)
- **Google Gemini**
- **Meta LLaMA**
- **Claude**

Applications of LLMs



- Conversational AI and chatbots
- Content writing and summarization
- Language translation
- Question answering systems
- Code generation

Types of AI Models



- Text Models – process language
- Image Models – analyze images
- Speech Models – recognize voice
- Embedding Models – convert data into vectors

Text Models (Natural Language Processing - NLP)



- These models understand, interpret, and generate text.
- **Large Language Models (LLMs):** These models, such as GPT-4 and Gemini, use transformer architectures. They predict the next part of a sequence to create text, code, or summaries.
- **Masked Language Models:** These models, like BERT, predict hidden words in sentences. They provide context for tasks such as classification or sentiment analysis.
- **Applications:** Chatbots, translation, content creation, and summarization

Image Models (Computer Vision)



- Image models process visual data to recognize, classify, or generate new visual content.
- **Generative Models (Diffusion/GANs):** Models like Stable Diffusion or DALL·E create new images from text prompts.
- **Convolutional Neural Networks (CNNs):** CNNs process grid-like data for image classification and object detection.
- **Applications:** Artistic generation, medical imaging analysis, and facial recognition

Speech Models (Audio Processing)



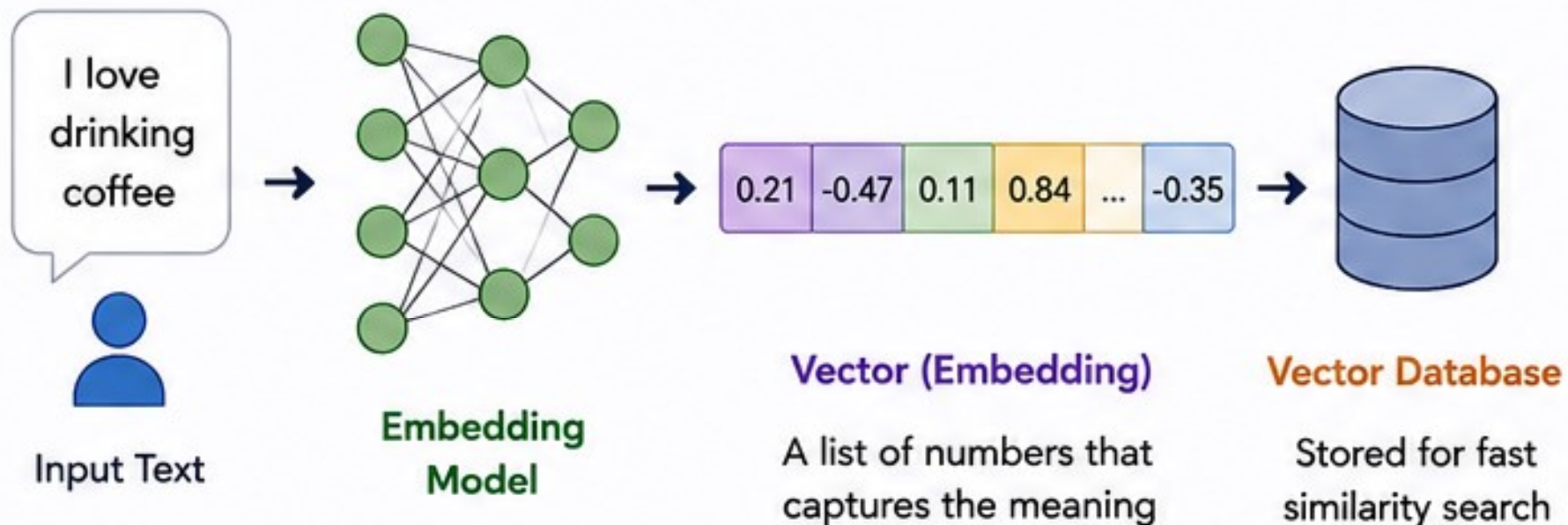
- Speech models connect spoken audio and text, often using two components.
- **Automatic Speech Recognition (ASR):** ASR converts spoken language into text (speech-to-text), often using encoder-decoder architectures.
- **Text-to-Speech (TTS):** TTS synthesizes human voices from written text.
- **Applications:** Virtual assistants (Siri, Alexa), transcription services, and live translation

Embedding Models (Vector Representations)



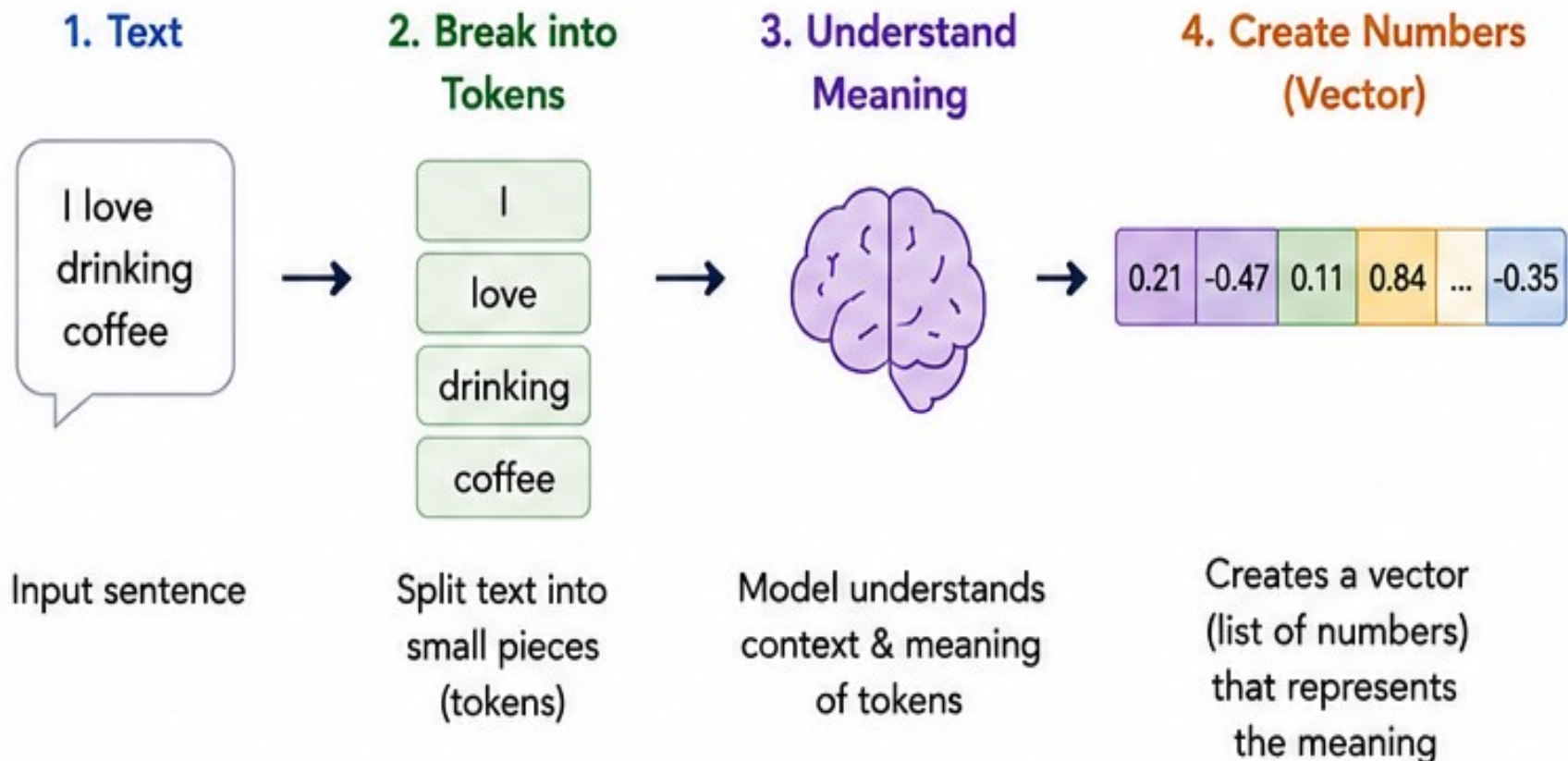
- Embedding models convert unstructured data (text, images, audio) into numerical vectors in a high-dimensional space.
- **Core Idea:** They capture the semantic meaning or features of data. Similar data points are placed closer together in this vector space.
- **Applications:** Semantic search, recommendation systems, and clustering

1. What is an Embedding Model?



Similar meanings → similar numbers (vectors).
Different meanings → different numbers.

2. How Text is Converted to Numbers?



3. How It Matches with Your Input?

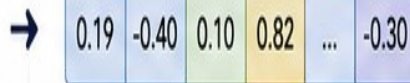
1. Your Search (Query)

Best coffee
near me



You enter a search
(query)

2. Convert to Vector



Your query is converted
into a vector

3. Compare with Database (Check Similarity)

I love drinking coffee		0.92	✓
Coffee is awesome		0.88	✓
Best coffee near me		0.95	★
I enjoy tea		0.20	✗

Similarity is measured (e.g., cosine similarity).
Higher score = more similar.

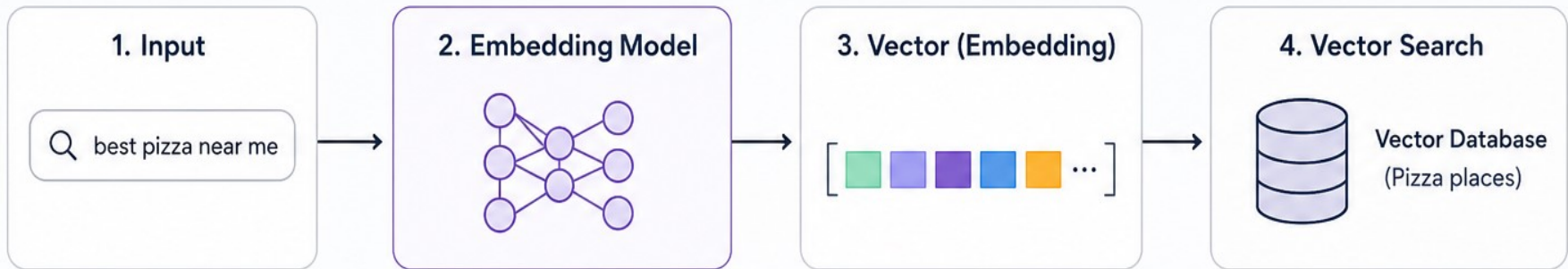
4. Return Top Matches

★ Best coffee near me

★ I love drinking coffee

★ Coffee is awesome

Top similar results
are returned



5. Top Results



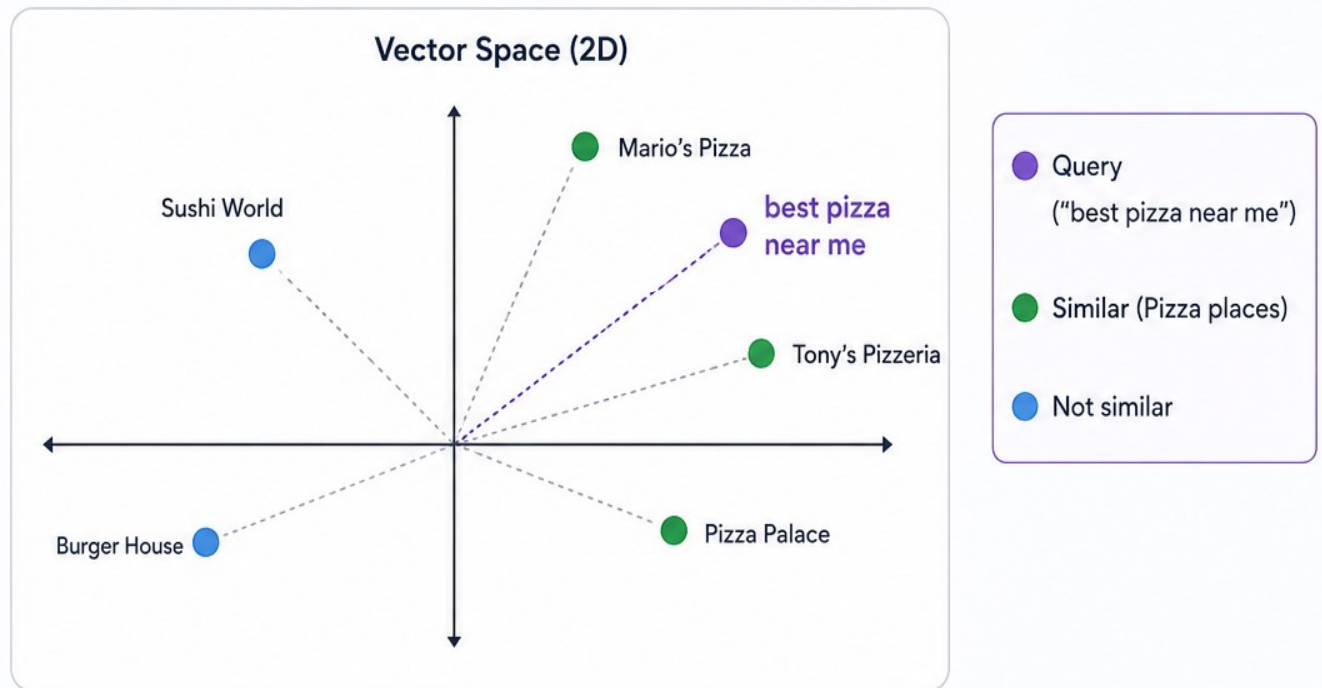
Mario's Pizza
★★★★★
0.12



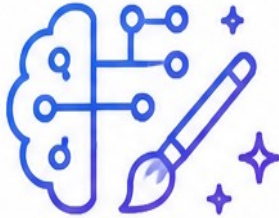
Tony's Pizzeria
★★★★☆
0.18



Pizza Palace
★★★★☆
0.25



Generative AI



Creates New Content

Generates original text, images, audio, code, and more.

VS

Analytical AI



Derives Insights from Data

Analyzes existing data to find patterns, trends, and insights.

	Creates something new	PRIMARY PURPOSE	Understands and interprets data	
	Creativity, generation, ideation	FOCUS	Analysis, prediction, decision support	
	Text, images, audio, video, code, designs	OUTPUT	Reports, insights, forecasts, recommendations	
	GANs, Diffusion Models, Transformers (e.g., GPT, DALL-E)	TECHNIQUES	Machine Learning, Statistical Models, Data Mining	
	Content creation, design, entertainment, automation	USE CASES	Business intelligence, risk analysis, fraud detection, forecasting	
	Expands possibilities, sparks innovation	VALUE DELIVERED	Optimizes decisions, drives efficiency	

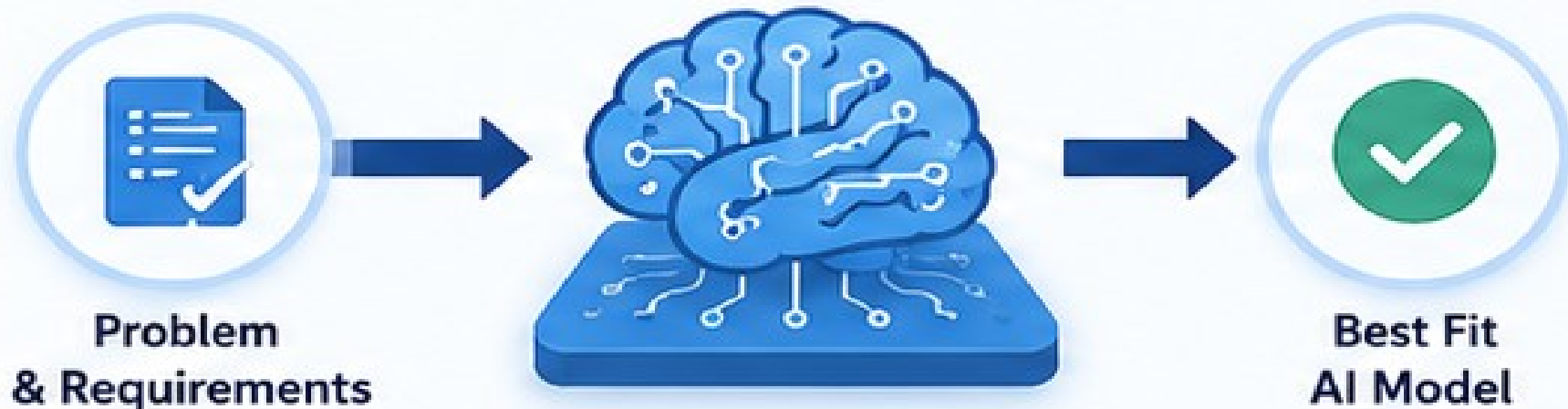
Exercises (Generative AI ,Analytical AI, Embedding Model)



- Which AI model is suitable for generating blog articles?
- Which AI model is best for detecting email spam?
- Which AI approach is suitable for recommendation systems?
- Which AI model is suitable for predicting house prices?
- Which AI model can create images from text prompts?
- Which AI model is used for customer support chatbots?
- Which AI model is best for classifying customer reviews as positive or negative?
- Which AI model is suitable for generating computer code?
- Which AI approach is used to compare the meaning of two sentences?
- Which AI approach is used to recommend similar products or movies?

AI Model Selection

AI model selection is the process of choosing the most suitable AI model for a specific product or application.



Different AI problems require different types of models depending on the type of data, task, and performance requirements.

Factors Affecting AI Model Selection

Factor	Description
Type of Problem	The model depends on whether the task involves text processing, image recognition, speech analysis, or prediction.
Type of Data	The format of data (text, image, audio, numerical data) determines which model should be used.
Accuracy Requirement	Some applications require very high accuracy, such as medical diagnosis.
Computational Resources	Powerful models may require high processing power and memory.
Speed and Cost	Real-time applications require faster and cost-efficient models.

Examples



**Chatbot
Application**



**Large Language
Models (LLMs)**



**Image
Recognition
System**



**Convolutional
Neural Networks
(CNN)**



**Speech
Recognition
System**



**Speech
Processing
Models**



**Recommendation
System**



**Machine Learning
Algorithms**





Single Agent AI ✨

A Single Agent AI is one AI system that **works alone** to complete tasks and make decisions.



Example: ChatGPT



One AI system understands the question and gives the answer.

More Real-Life Examples of Single Agent AI

Siri / Alexa

What's the weather today?



One AI assistant listens to your voice and answers you.

Netflix Recommendations



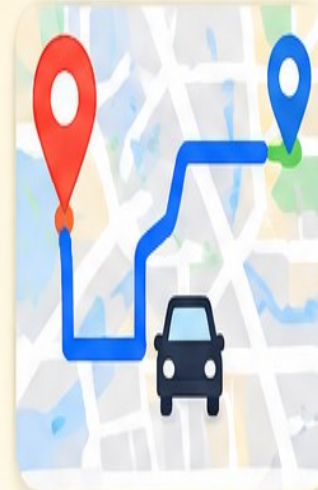
One AI system suggests movies/shows based on your likes.

Face Unlock (on Phone)



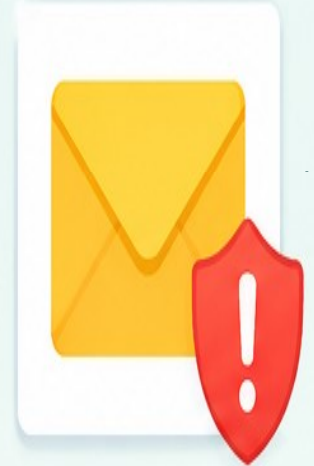
One AI model recognizes your face and unlocks your phone.

Google Maps (Route Suggestion)



One AI system finds the best route for you.

Spam Filter (in Email)

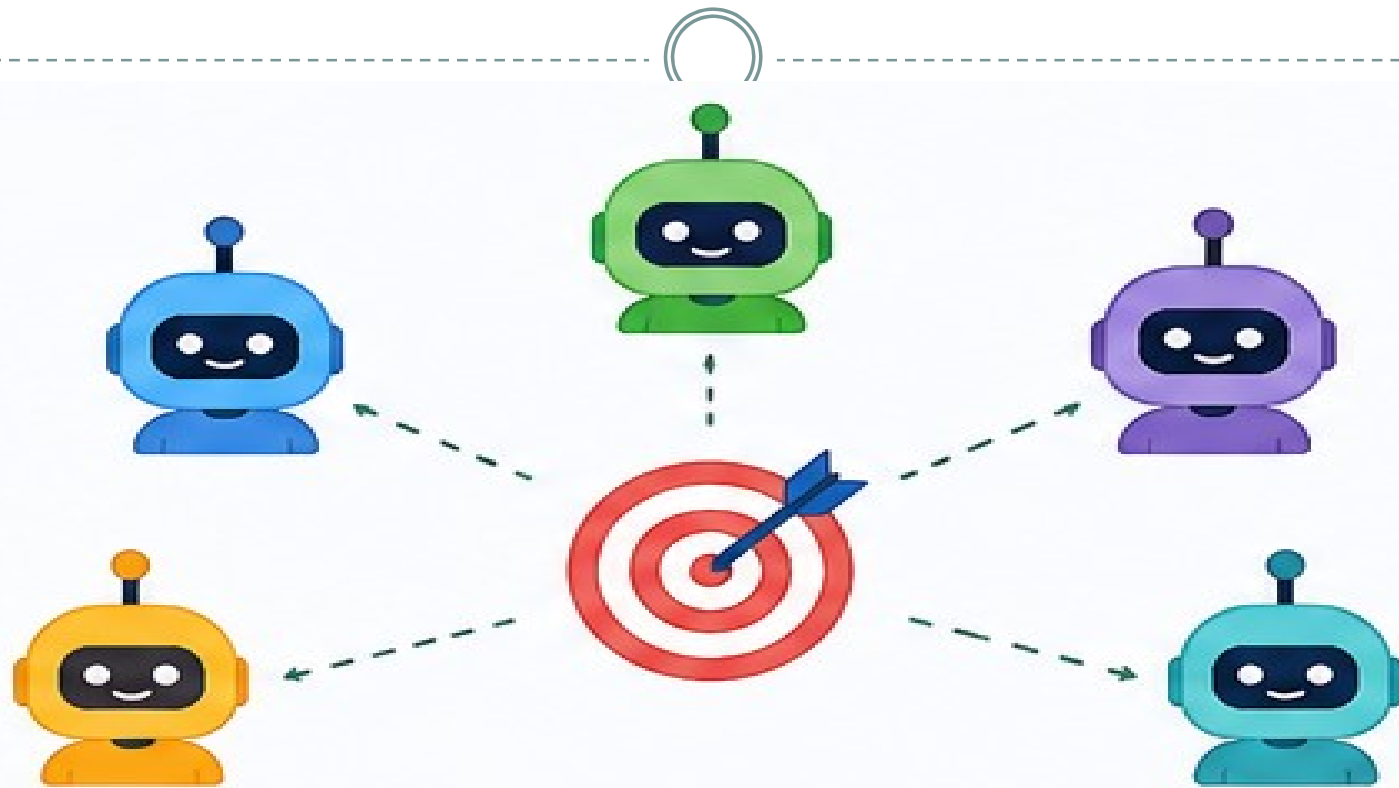


One AI model detects and filters spam emails.

Multi-Agent AI



- Multiple AI agents working together
- Each agent performs a specific task
- Improves efficiency and automation



Multi-Agent AI means many AI agents work together to achieve a **common goal**.

How does it work?

1. Task



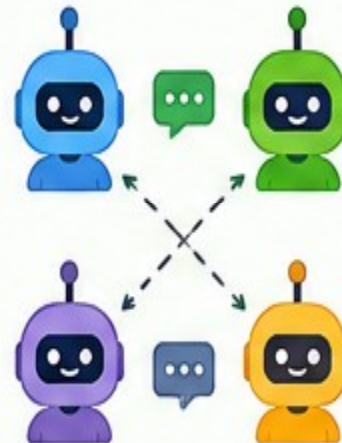
A big task is given to the system

2. Agents



The task is shared among multiple agents

3. Work Together



Agents communicate, share information and cooperate

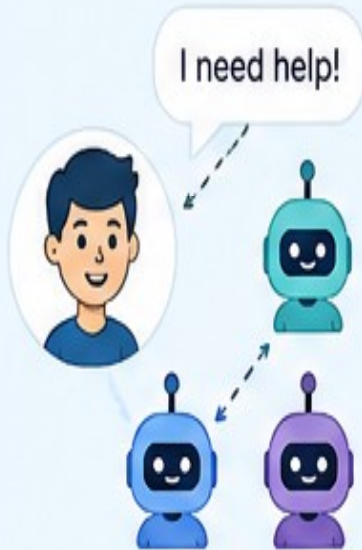
4. Complete Task



Together they complete the task better and faster

Examples

Chatbot Team



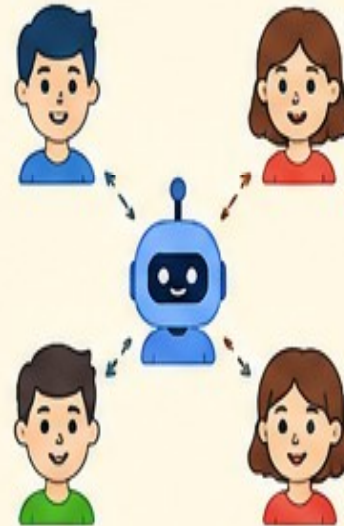
Different agents handle greetings, answers, and follow-ups.

Smart School



Agents manage different school tasks together.

Group Project



Like students in a group, agents divide work and help each other.

Game Team



Agents play different roles and work as a winning team.

Types of Multi-Agent Systems

1. Cooperative MAS



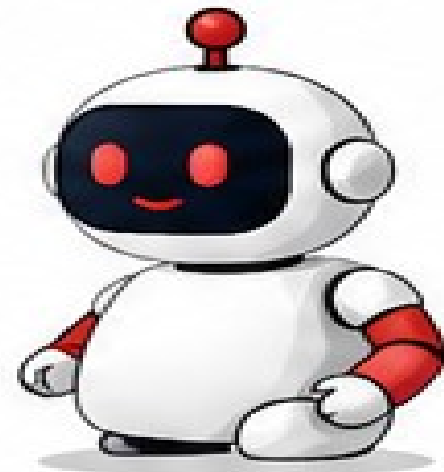
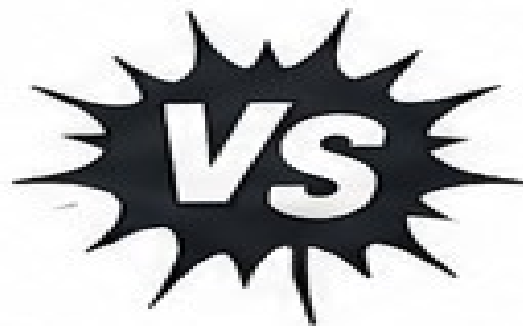
Description:

Agents work together to achieve a common goal.

Example:

Team of robots exploring a disaster area.

2. Competitive MAS



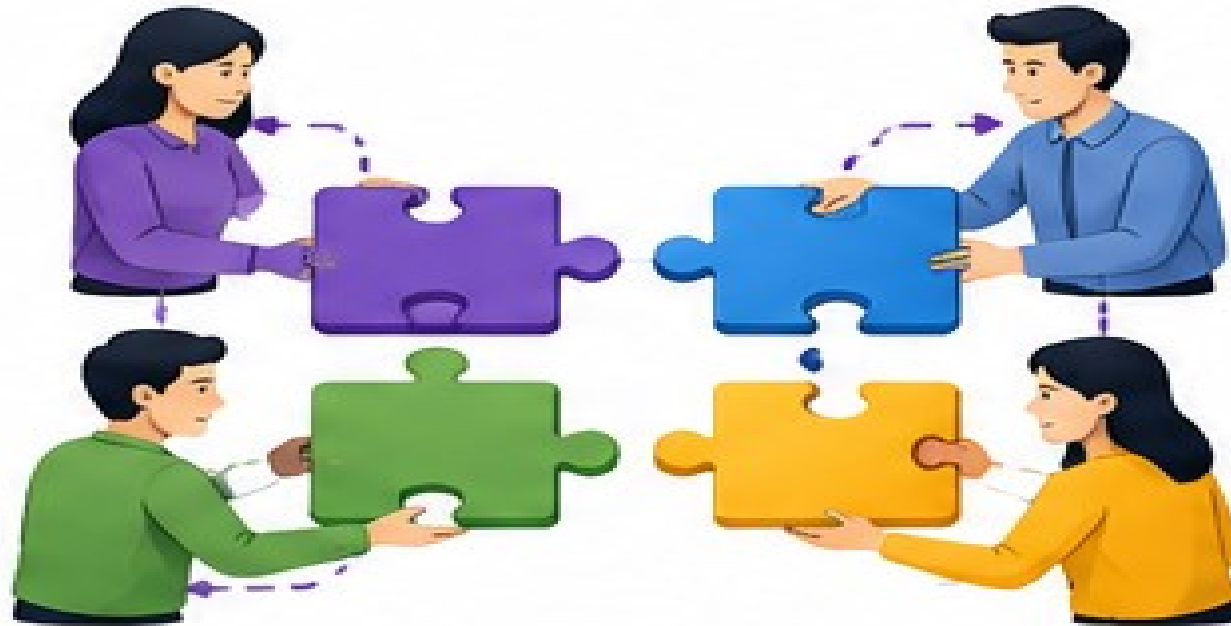
Description:

Agents have conflicting goals and compete for resources or outcomes.

Example:

Bidding agents in an online auction.

3. Collaborative MAS



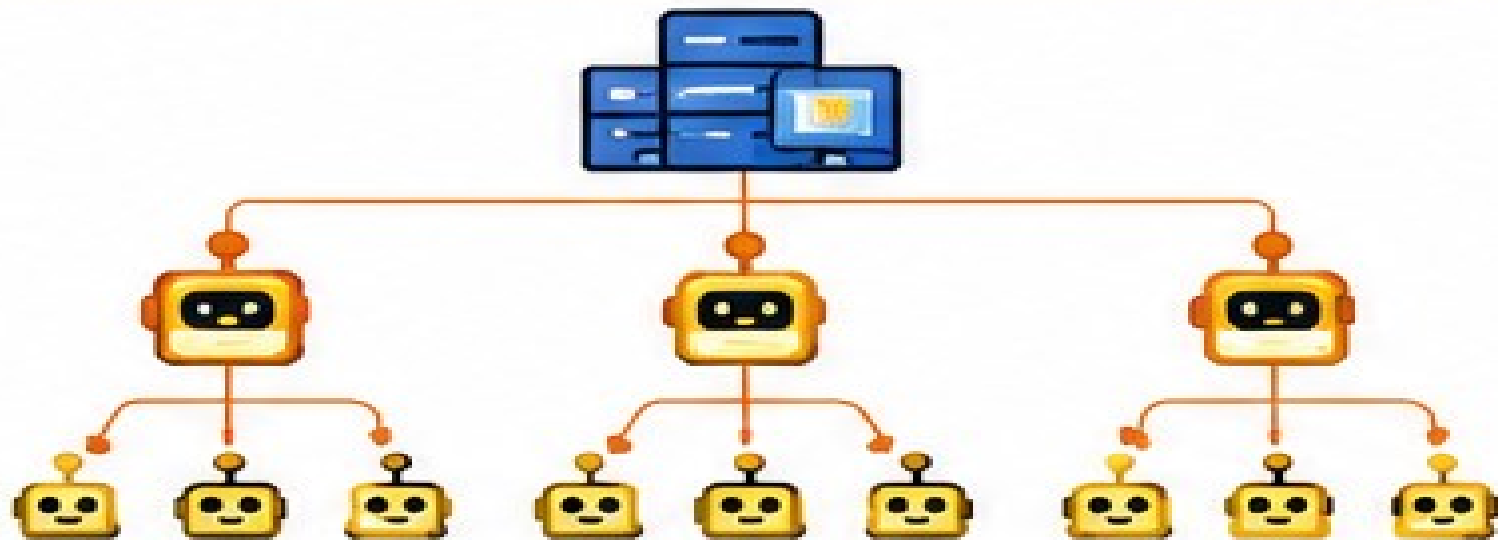
Description:

Agents collaborate for mutual benefit while having individual goals.

Example:

Supply chain management with partner organizations.

4. Hierarchical MAS



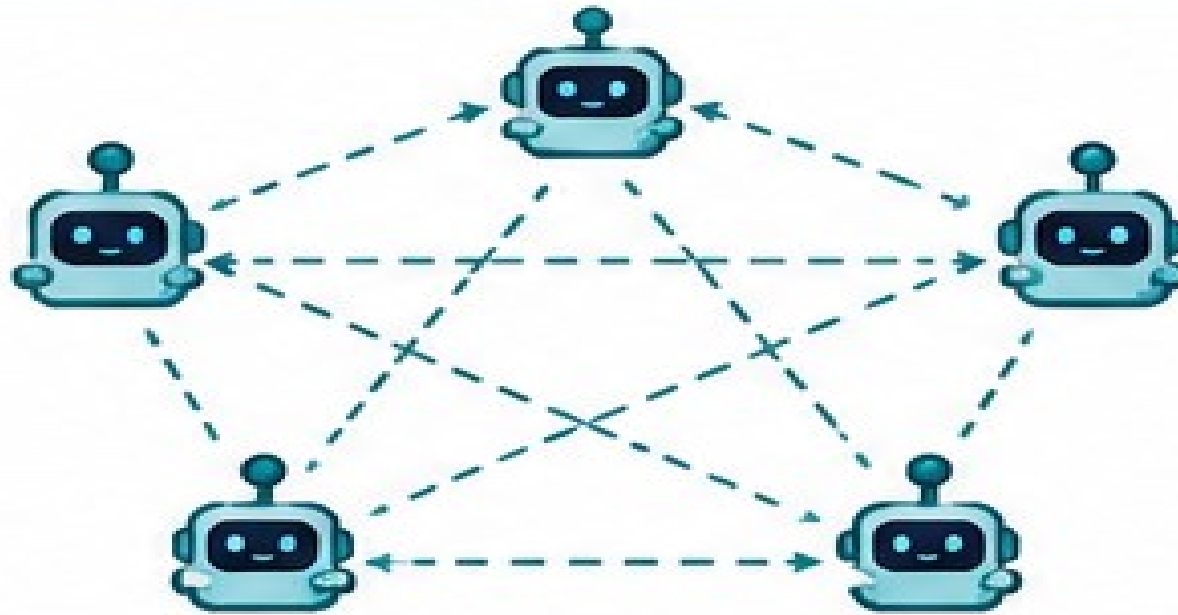
Description:

Agents are organized in a hierarchy with different levels of authority.

Example:

Command and control systems in the military.

5. Distributed (Decentralized) MAS



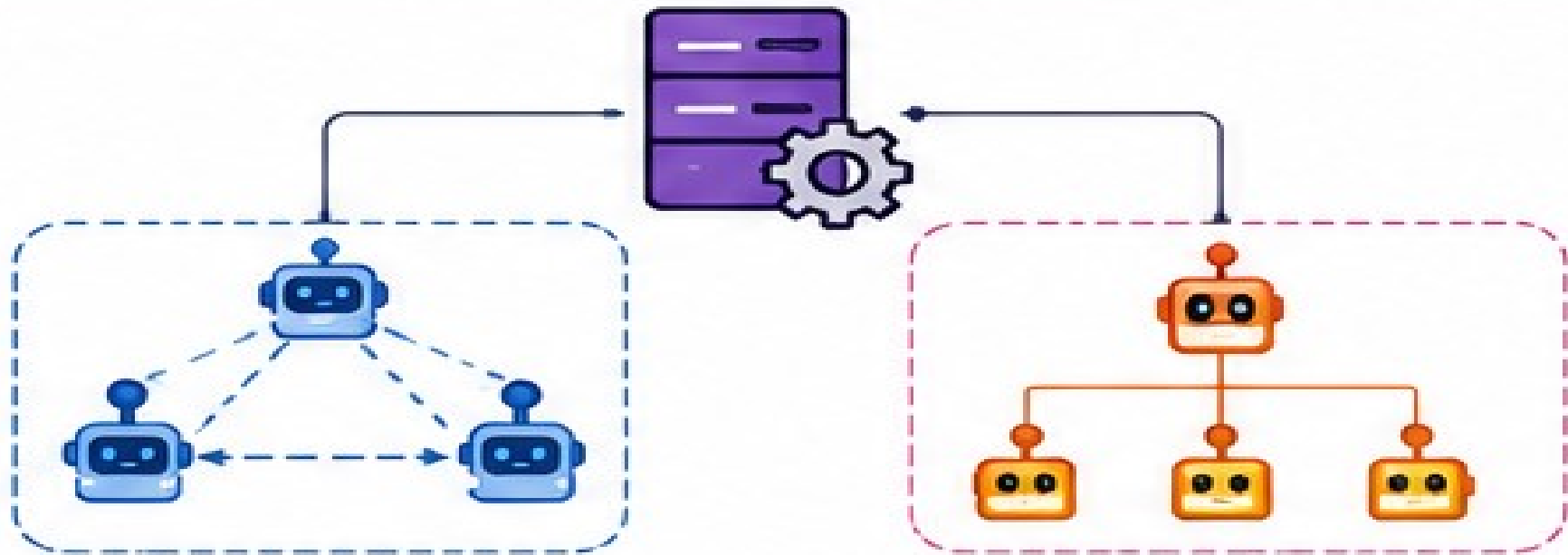
Description:

No central control; agents interact and make decisions autonomously.

Example:

Peer-to-peer file sharing networks.

6. Hybrid MAS



Description:

Combines two or more types of MAS to solve complex real-world problems.

Example:

Smart city systems combining hierarchical and distributed approaches.

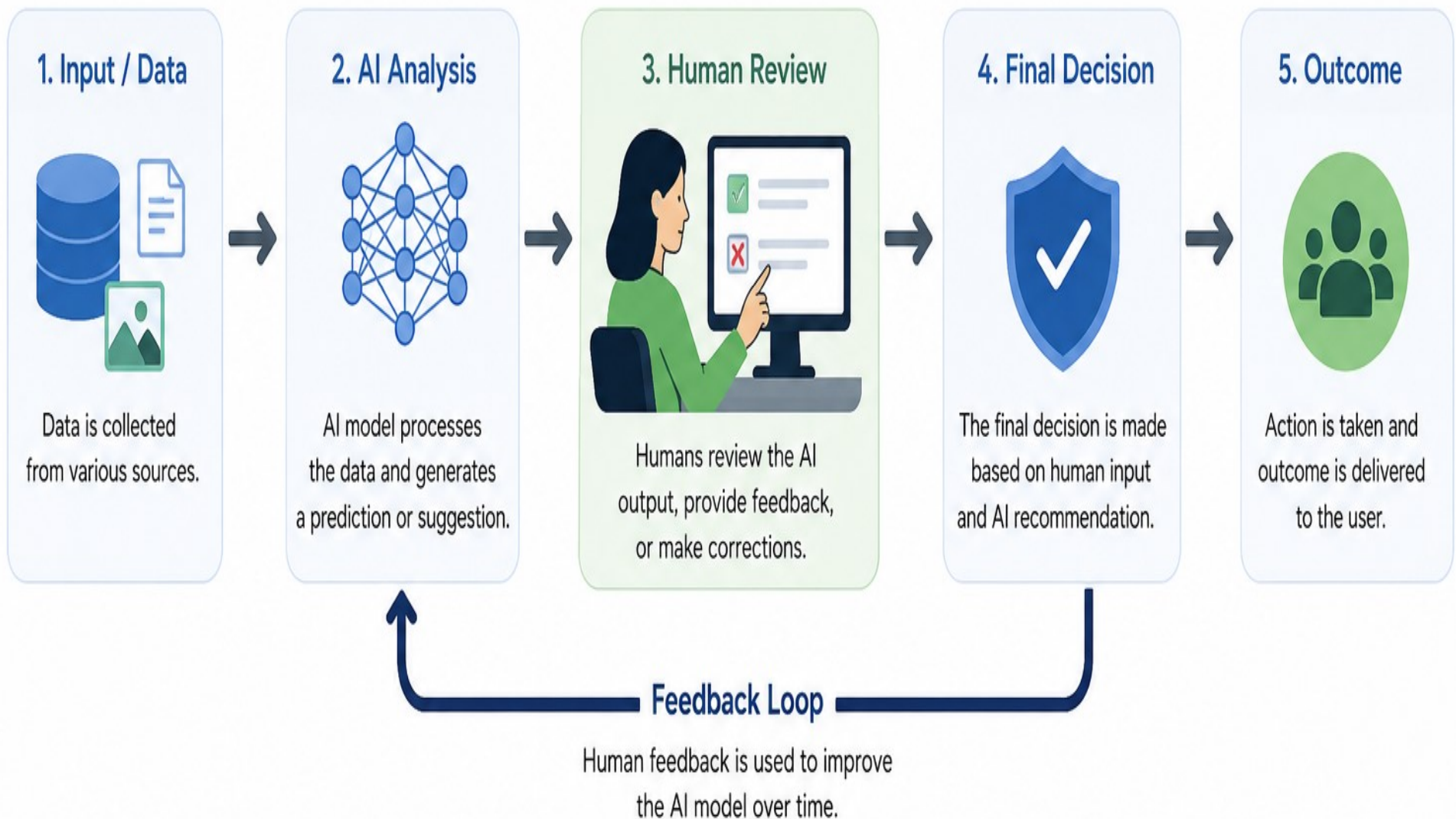
Human-in-the-Loop Systems



- Human-in-the-Loop (HITL) systems are AI systems in which **humans participate in the decision-making process along with Artificial Intelligence.**
- Ensures safety and reliability
- Used in healthcare, finance, moderation

Human-in-the-Loop (HITL) AI Systems

AI systems in which humans participate in the decision-making process along with Artificial Intelligence.



Key Features of Human-in-the-Loop Systems



Feature	Description
Human Supervision	Humans monitor AI outputs and verify results.
Error Correction	Humans can correct mistakes made by AI models.
Continuous Learning	AI systems improve using human feedback.
Better Accuracy	Combining human intelligence with AI improves reliability.

Where is HITL AI Used?

AI + Human Judgment = Better Decisions



1. Healthcare

AI suggests diagnoses; doctors review and confirm treatment.



2. Finance

AI detects risky transactions; analysts review and approve actions.



3. Content Moderation

AI flags harmful content; moderators review and take appropriate action.



4. Autonomous Systems

AI navigates and makes decisions; humans supervise and take over when needed.



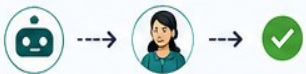
5. Legal & Compliance

AI analyzes documents; lawyers review and decide the outcome.



6. Customer Support

AI handles common queries; human agents solve complex or sensitive cases.



7. Education

AI personalizes learning and grades; teachers review and guide students.



8. Manufacturing & Quality Control

AI detects defects; inspectors verify and ensure quality.



9. HR & Recruitment

AI screens and ranks candidates; recruiters interview and hire.



10. Scientific Research

AI analyzes data and finds patterns; researchers validate and publish results.



AI

Processes data fast and at scale



Human

Adds judgment, context and ethics



Better Outcomes

Safer • Fairer • More Reliable
Trusted • Responsible



More Accurate

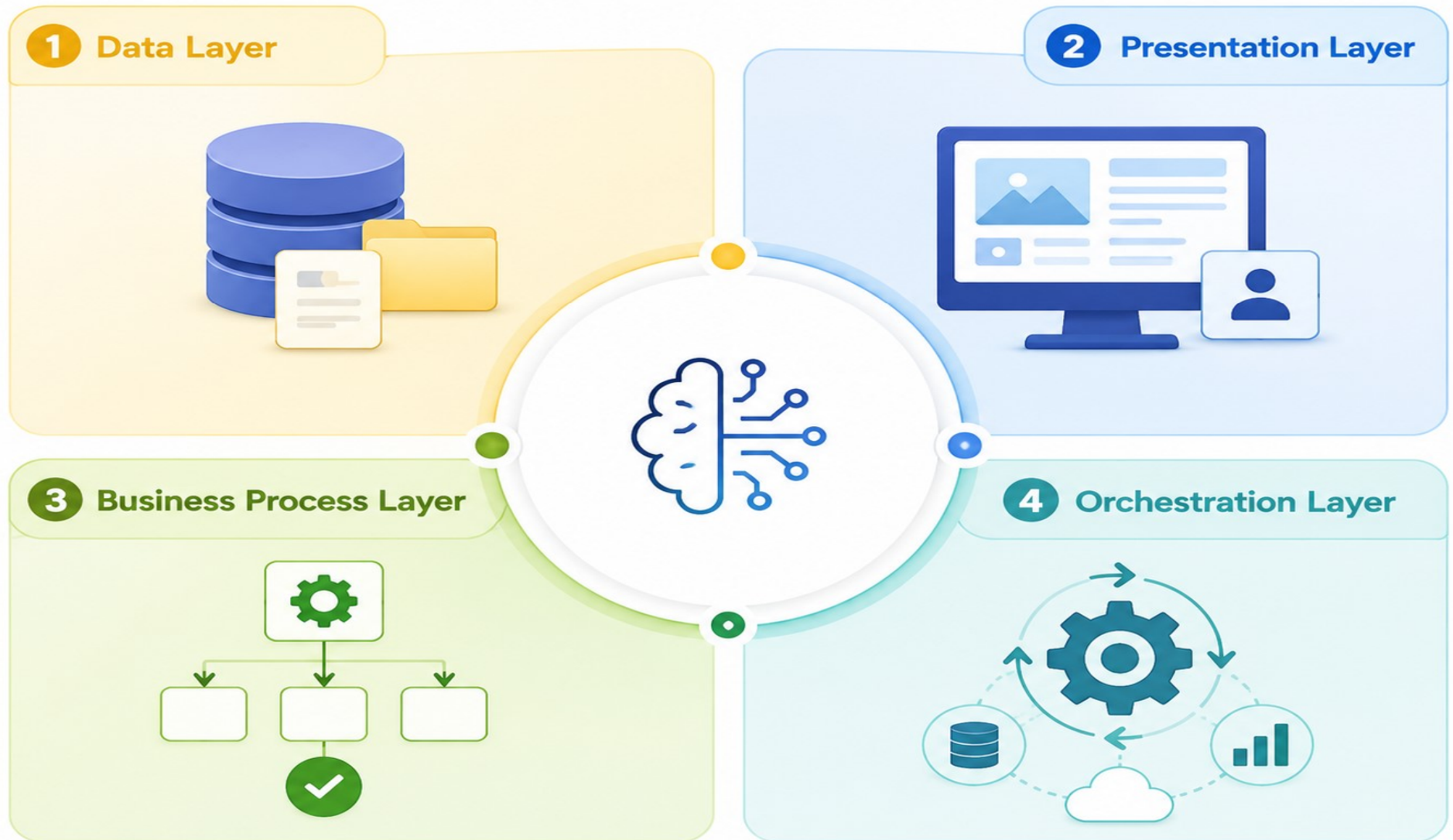


Reduced Risk



Stronger Trust

Basic AI System Architecture

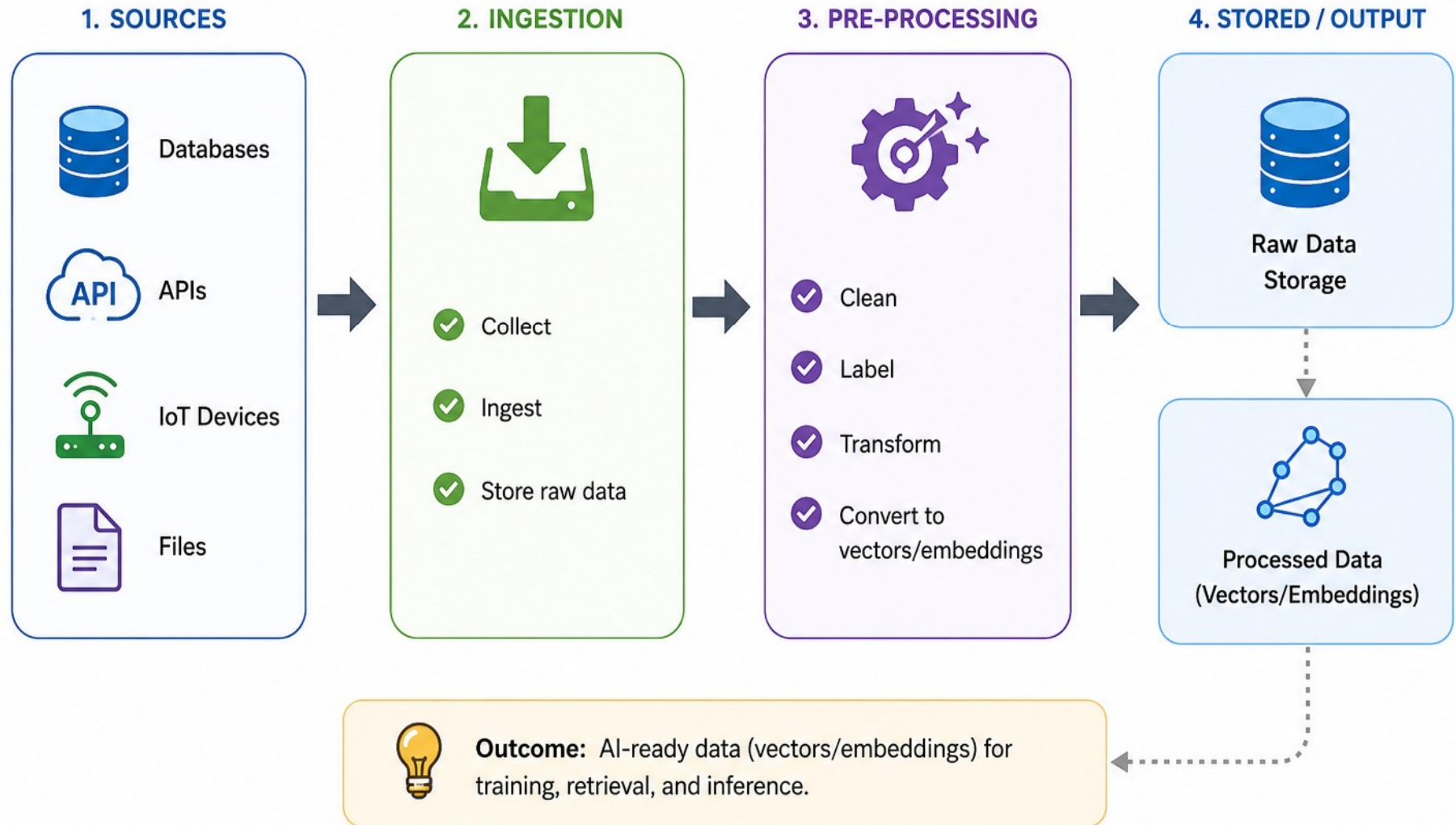




DATA LAYER

Gathers and stores data from various sources.

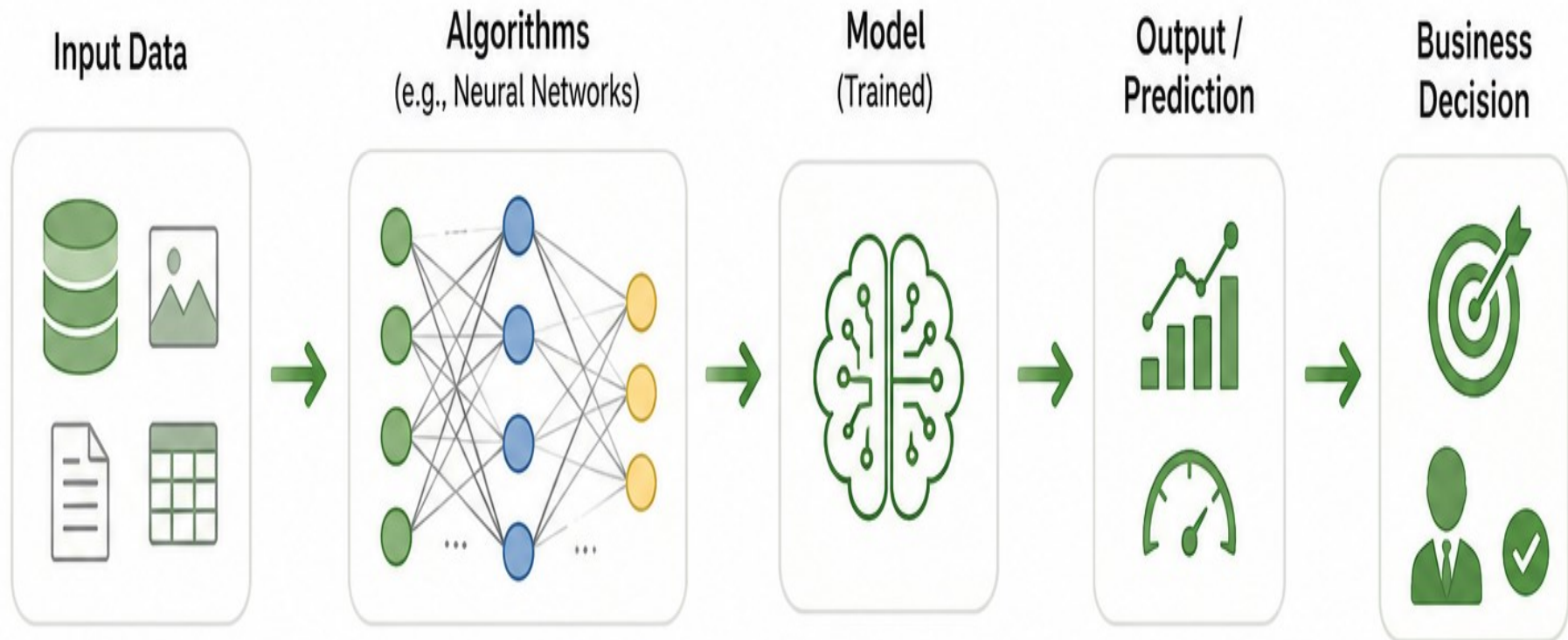
Includes **data ingestion** and **pre-processing** to convert raw data into **vectors/embeddings** that AI can understand.

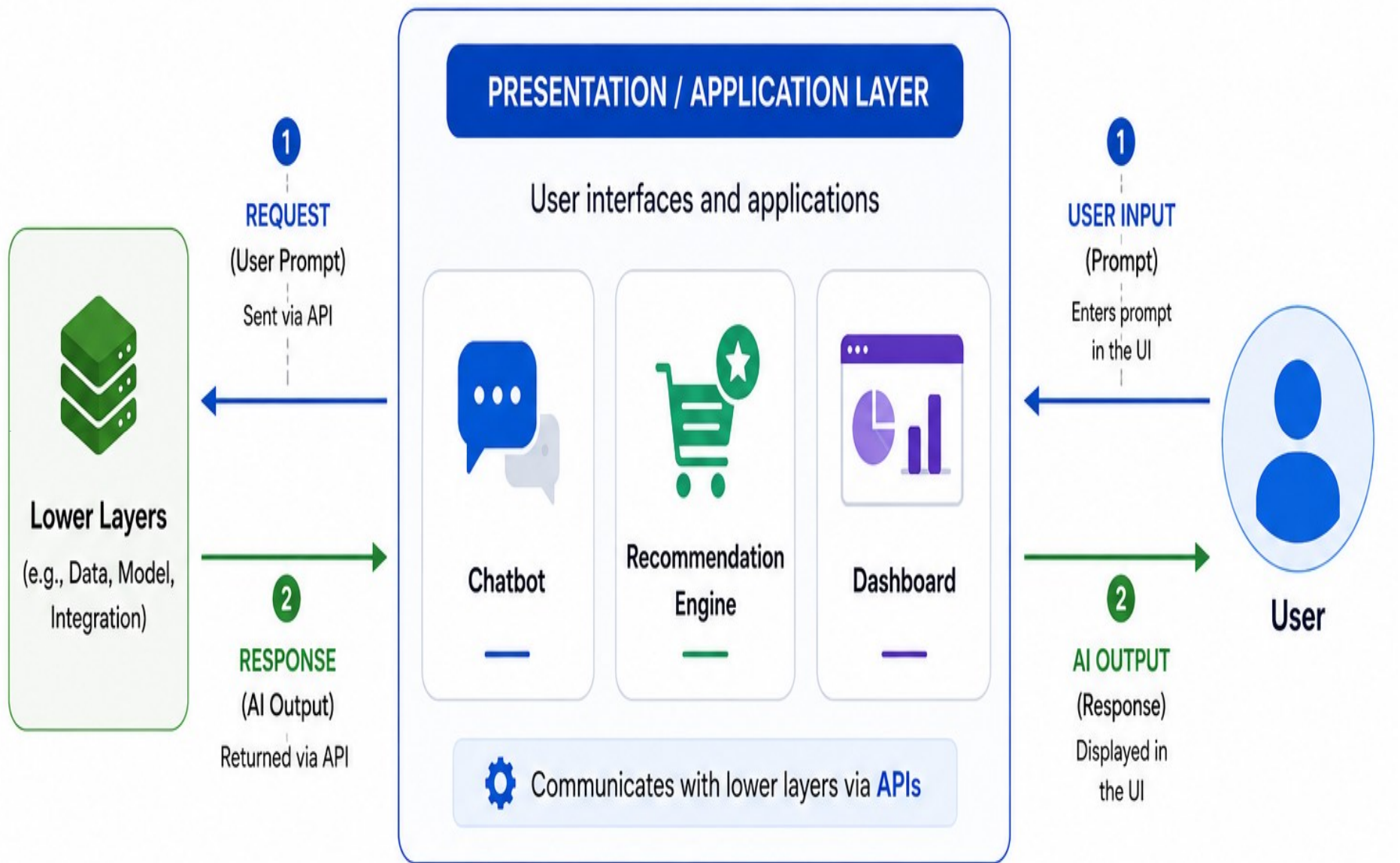




Model / Business Process Layer

This is where the actual machine learning happens.
It consists of the **Algorithms** (like Neural Networks) and the **Models** themselves.





Orchestration & Infrastructure Layer

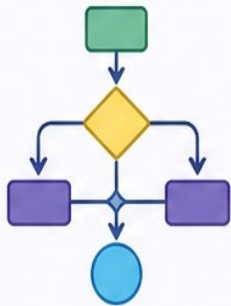


Provides computing power (GPUs, CPUs) and tools to run AI.



Manages models, monitors performance and scales as users grow.

Run Workflows



Manage Models



Monitor



Scale Automatically



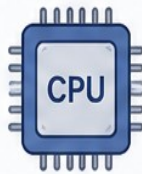
Secure & Govern



Infrastructure Resources



GPUs



CPUs



Storage



Cloud Services



Networking



Security

3–4 Mark Questions



- What is an AI product?
- Differentiate between AI tool and AI product.
- List out components of AI systems and explain any two.
- Define Generative AI. Write Applications of Generative AI.
- What are Large Language Models (LLMs)?
- Explain text/image/speech/embedding AI model with application.
- Differentiate Generative AI and Analytical AI.
- Give an example of Multi Agent AI.
- Short Note on :Human-in-the-loop systems.

7 Mark Questions



- Explain components of AI systems with examples.
- Describe Generative AI with its applications and examples.
- Discuss the working of a Large Language Model (LLM) and its applications.
- Describe the factors affecting for AI model selection
- Explain Multi agent AI and its types.
- Illustrate the basic AI system architecture with a neat diagram and explain each layer.